

Prediction of ‘awakening’ and outcome in prolonged acute coma from severe traumatic brain injury: evidence for validity of short latency SEPs

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Abstract

Objective: To evaluate the prognostic value of somatosensory evoked potentials (SEPs) in severe traumatic brain injury (TBI) considering both ‘awakening’ and disability.

Methods: SEPs were recorded in 60 severe TBI with duration of acute coma > 7 days. N20–P25 amplitudes, their side-to-side asymmetry and CCT were measured. SEPs on each hemisphere were classified as normal (N), pathological (P) or absent (A). ‘Awakening’ and disability were assessed after at least 12 months using Glasgow Outcome Scale (GOS). SEP predictive value was compared with GCS and EEG reactivity

Results: Seventy-five percent regained consciousness. 29/60 had a good outcome (GOS 4–5) and 31/60 had a bad outcome (GOS 1–3). According to the ROC curve, SEP findings were classified in 3 grades. Grade I (NN, NP) had PPV of 93.1% for ‘awakening’ and 86.2% for good outcome. Grade III (AA) had PPV of 100% for bad outcome and 72.7% for ‘awakening’. Grade II (PP, NA, PA) was associated with the wider range of outcome. A multivariate analysis including SEP grading, GCS and EEG reactivity did not increase the percentage of cases prognosticated by SEP alone.

Conclusions: We confirm the high predictive value of SEPs in TBI, which is greater than GCS and EEG reactivity. Indeed, SEP grades I and III were able to predict the correct prognosis in more than 80% of severe TBI. Therefore, SEPs should be used more widely in the prognosis of severe TBI.

Significance: Differently from post-anoxic, in post-traumatic coma the presence of normal SEPs has a favourable predictive value both for ‘awakening’ and disability. We think that in literature enough attention has still not been paid to this finding.

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Keywords: Somatosensory evoked potentials; Coma; Traumatic brain injury; Outcome prediction; EEG; GCS

1. Introduction

Glasgow Coma Scale (GCS) and EEG are widely used in the intensive care unit to evaluate comatose patients. Yet, their prognostic utility is dependent on the etiology of the coma. An extensive review of the literature reveals that neither GCS nor EEG are predictors as sensitive in

traumatic coma as in hypoxic–ischemic coma (Attia and Cook, 1998; Facco and Munari, 2000), thus suggesting there may be a greater role for ancillary Evoked Potential (EP) testing. Among EPs, somatosensory evoked potentials (SEPs) provide a powerful instrument to evaluate brain function in head injury because of the extensive neural pathways involved. Ideally, SEPs should be a reliable outcome predictor for as many patients as possible belonging to different outcome categories. Most of the meta-analytic studies on traumatic coma (Attia and Cook, 1998; Carter and Butt, 2001; Rothstein, 2000), have focused

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on SEP findings only to predict poor prognosis. Actually, primary bilateral loss of cortical responses is regarded as one of the best prognostic indicators predicting, with very few exceptions, death, permanent vegetative state, or, at best, severe disability. According to the two main meta-analysis on the topic (Robinson et al., 2003; Rothstein, 2000) the percentage of patients with severe traumatic brain injury (TBI) who die or remain in a vegetative state is lower (39–44%) than those with anoxic–ischemic coma (71–77%). Therefore, in post-traumatic coma, more attention should be paid to which SEP patterns are reliable predictors for other outcome categories such as good recovery and mild disability.

The patients who, after one week, have already clinically evolved ('awakened' or died) have a limited benefit from early SEP recording. It is more difficult, instead, to establish the prognosis for those patients with a prolonged state of acute coma (> 1 week) and with a scanty clinical evolution within the first week. Based on these considerations, we have decided to reconsider the topic evaluating the capacity of short-latency cortical SEPs to predict the functional outcome, defined after 1 year, in this subcategory of severe TBI patients.

The aims of our study were two-fold. Firstly, we correlated SEP findings with the outcome (as described in previous studies), considering both favourable and unfavourable patterns (different SEP gradings) at two levels of outcome categorization ('awakening' and disability). Secondly, we compared SEP prognostic value with the GCS score and EEG reactivity.

2. Methods and materials

We reviewed retrospectively the patients with severe post-traumatic coma, admitted in the Intensive Care Unit (ICU) of Careggi Hospital between 1996 and 2001.

Inclusion criteria were: (a) severe head trauma, GCS on admission ≤ 7 ; (b) prolonged state of acute coma (> 7 days); and (c) lack of pre-existing brain diseases or impairment of peripheral and spinal SEPs.

We identified a group of 60 adult patients (mean age 27, range 17–72); neurosurgery was carried out in 23 patients (36.6%), before SEP recordings, whereas the remaining 37 patients underwent medical management according to the standard protocol of the ICU. We did not have at our disposal an index of hypoxia which eventually occurred at an early stage of trauma before admission.

The clinical conditions of these patients changed very little during the first week: mean GCS was 4.7 on admission (before sedation and muscle relaxation) and 4.9 after one week. The mean duration of acute coma (time required for patient's eyes to open, spontaneously or after verbal stimuli) was 14 days (range 8–56).

EEGs were recorded with an 8-channel ERA-9 using stainless steel needle electrodes placed according to

the international 10–20 system (Fp1–C3, C3–T3, T3–O1, Fp2–C4, C4–T4, T4–O2). The EEG was performed after 48–72 h and after 1 week (mean 10 days). Sedation, if present, was interrupted more than 6 h before the recording. EEG duration was at least 30 min. Reactivity was tested by shouting the patient's name, applying painful stimuli (pressure to the fingernail and infraorbital foramen) and passively opening/closing the eyes.

Two of the authors (AA, AR), unaware of the patient's outcome, evaluated the EEGs. Failing a universally accepted EEG classification in post-traumatic coma and taking into account previous data (Gütling et al., 1995), we decided to consider only the EEG reactivity.

EEG reactivity was rated according to two different categories: (1) present (amplitude and/or frequency changes) and (2) absent.

SEPs were recorded 10–13 days (mean 11.3) after the injury. Electrical stimulation of the right and left median nerve was delivered to the wrist by a bipolar surface electrode with intensity of 4–5 mA above the motor threshold. A 4-channel EP apparatus was used. Pulse duration was 0.2 ms; stimulus rate 3 Hz; time base 100 ms; bandwidth 5 Hz–3 KHz. Recording stainless steel needle electrodes were placed at Erb's point (referred to contralateral Erb's point), spinous process Cv7 (referred to anterior neck), C3' and C4' (referred to Fz and ipsilateral mastoid). At least two averages of 300 responses were repeated and superimposed. N9, N13 and N20 latencies were measured. We determined peak-to-peak amplitude of N20/P25 using Fz reference; even if Fz reference produces a larger inter-individual amplitude variability, it avoids the influence of subcortical potentials and reduces noise level. Moreover, Fz reference allows the recognition of cortical responses even when their presence is doubtful looking at traces with mastoid or earlobe reference. P25 was identified as the major positive peak following N20. The central conduction time (CCT), defined as the interpeak latency N13–N20, was considered pathological if longer than 7.8 ms (mean + 3 SD of our laboratory standards: $5.9 \text{ ms} \pm 0.63$). For the large inter-individual variability of N20/P25 amplitude, we used the percentiles to determine the lowest limit of normality ($1.2 \mu\text{V} = 5^\circ$ percentile).

SEPs of each side were graded on a 3-point scale in blind fashion by two of the authors (AA, AG) considering the amplitude of cortical complex (N20/P25) and CCT. SEPs were graded as normal (N) if N20/P25 amplitude and CCT were normal; pathological; (P) if CCT was abnormally prolonged and/or N20/P25 amplitude was $< 1.2 \mu\text{V}$ or the left–right amplitude asymmetry was greater than 50%; absent (A) if cortical responses were absent with preserved cervical N13 (Fig. 1). We evaluated the absence of intracortical SEPs considering both Fz and mastoid references. In some patients, because of a severe axonal injury, we observed an extreme reduction of N20 or N20–P25, but the persistence of reproducible subsequent components in the range of 50–100 ms

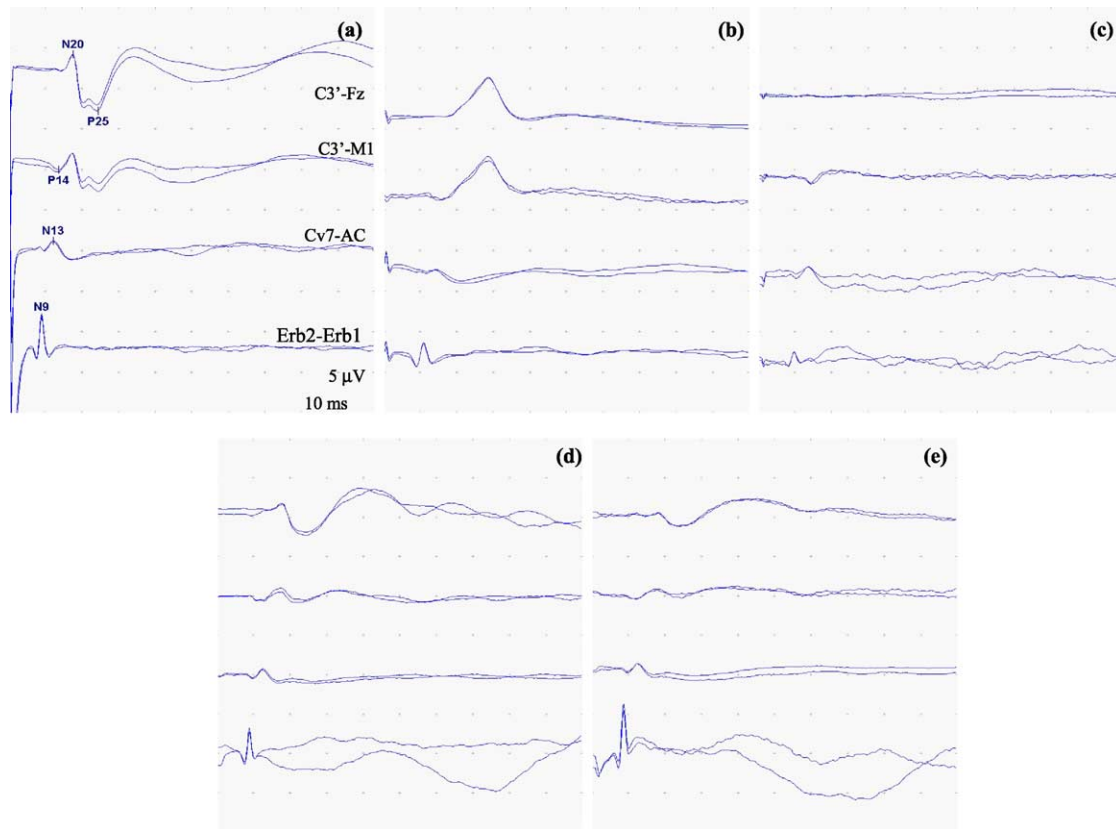


Fig. 1. Examples of SEP grading classification: (a) normal SEP (N) after stimulation of Median nerve; (b) pathological SEP (P) for an increased CCT; (c) absent cortical SEP (A) considering both Fz and mastoid references; (d) and (e) pathological amplitude asymmetry (>50%) of cortical SEPs between the two hemispheres (P) with normal CCT. (Calibration, time base, montages and component identification as in (a)).

post-stimulus interval. That is the reason why we evaluated the absence of intracortical SEPs along the whole time interval of 100 ms.

Taking into account responses in both hemispheres, 6 SEP patterns were defined: NN, NP, PP, AN, AP, and AA.

A follow-up telephone interview was conducted by two of the authors (CC, AP) at least 1 year after discharge from the hospital. The outcome was assessed using the 5-point Glasgow Outcome Scale (GOS, Jennett and Bond, 1975): death (1); vegetative state (2); severe disability (3); moderate disability (4); and good recovery (5). GOS levels were merged into two dichotomic categories: (a) 'awakening' (GOS 3–5) and death/permanent vegetative state (PVS, GOS 1–2); (b) good outcome (GOS 4–5) and bad outcome (GOS 1–3).

Statistical analysis was performed using the StatView Software Package (SAS Institute). The accuracy of SEPs in predicting the outcome was first evaluated by plotting receiver operating characteristic (ROC) curves, which combine both the sensitivity and specificity achieved by the diagnostic test at several cut-off points (Fletcher et al., 1996). Then, in order to achieve an adequate number of cases in each grading set for subsequent univariate and multivariate analysis, we

collapsed SEP patterns in a 3-grade classification scheme looking at ROC curves.

Logistic regression analysis was used to calculate the percentage of correct and incorrect predictions of both good and bad outcome for GCS, EEG reactivity, and SEP grade separately. We computed sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV). We calculated also the likelihood ratio (LR) (~95% confidence interval, CI), with

Table 1
Patients with different outcome, grouped according to SEP classification

SEP Pattern	GOS Score				
	Good outcome		Bad outcome		
	No or mild disability	Moderate disability	Severe disability	Vegetative state	Death
NN (n=14)	4	9	0	0	1
NP (n=15)	3	9	2	0	1
PP (n=11)	1	2	5	1	2
NA (n=2)	0	0	1	0	1
PA (n=7)	0	1	5	0	1
AA (n=11)	0	0	3	4	4

N, normal; P, pathological; and A, absent.

Table 2

Accuracy of 'awakening' prediction using the most powerful SEP category classification, GCS and EEG reactivity

	Predicted outcome		Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	LR+ (CI)
	'Awakening'	Death/PSV					
<i>GCS</i>							
≥5	33	5	73.3	66.7	86.8	45.5	2.2 (1.0–4.5)
<5	12	10					
<i>EEG reactivity</i>							
Yes	30	6	66.7	60.0	83.3	37.5	1.6 (0.8–3.2)
No	15	9					
<i>SEP grading</i>							
I (NN, NP)	27	2	60	86.7	93.1	41.9	4.5 (1.2–16.7)
II (PP, NA, AP)	15	5	33.3	66.7	75.0	25.0	1 (0.4–2.2)
III (AA)	3	8	53.3	93.3	72.7	85.7	8.0 (2.4–26.3)

PPV, positive predictive value; NPV, negative predictive value; LR+, likelihood ratio; and CI, confidence interval at 95% (in bold type cases correctly classified).

the advantage of taking into account both specificity and sensitivity.

In addition, a multivariate analysis was also used to verify whether the combination of GCS, EEG reactivity and SEP grade improved the outcome prediction.

3. Results

Forty-five patients (75%) regained consciousness, 10 died and 5 remained in PVS Overall, a good outcome (GOS 4–5) was observed in 48% and a bad outcome (GOS 1–3) in 52%. Age and surgery did not correlate significantly with the GOS score.

Table 1 shows the patients with different outcomes, grouped according to SEP classification. Twenty-nine patients had bilaterally or monolaterally normal SEPs: 27 'awoke', 25 of them with a good outcome. All the 11 patients with bilateral loss of cortical responses had a bad outcome. Nine patients had unilateral absence of cortical SEPs, 8 of them were left with severe disability or died. Finally, bilaterally preserved, but abnormal SEPs was the category with the greatest prognostic uncertainty.

GCS ≥ 5 on the 10th day was a good predictor of 'awakening' (33/38), but not as good in predicting functional recovery (9 out of the 38 patients were left with severe disability). Seventeen out of the 22 patients with GCS < 5 had a bad outcome, while 5 had a good outcome. Thirty-six patients had a reactive EEG on the 10th day: 30 regained consciousness, but 7 of them survived with severe disability. Among the 24 patients with unreactive EEGs, 18 had a bad outcome (half of them with severe disability), but a relevant 25% (6/24) had good functional recovery. (Tables 2 and 3).

ROC curves, based on sensitivity and specificity of SEP patterns in the prediction of 'awakening' and disability, are shown in Fig. 2. The areas under the ROC curves, which estimate quantitatively the test performance, were 0.69 (CI 0.53–0.86) for 'awakening' and 0.91 (CI 0.84–0.98) for the level of disability expressing the higher prediction value of SEPs for the latter. The most powerful aggregation of SEP patterns resulted to be the following 3-graded scale: grade I (NN, NP), grade II (PP, NA, AP) and grade III (AA).

Tables 2 and 3 compare the results of univariate analysis of each predictor (GCS, EEG reactivity, SEP grade).

SEP grade I has a similar specificity (86.7 and 87.1%, respectively) and very high PPV (93.1 and 86.2%) in

Table 3

Accuracy of outcome prediction using the most powerful SEP category classification, GCS and EEG reactivity

	Predicted outcome		Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	LR+ (CI)
	Good (4–5)	Bad (1–3)					
<i>GCS</i>							
≥5	24	14	82.8	54.8	63.2	77.3	1.8 (1.2–2.7)
<5	5	17					
<i>EEG reactivity</i>							
Yes	23	13	79.3	58.1	63.9	75.0	1.8 (1.2–2.9)
No	6	18					
<i>SEP grading</i>							
I (NN, NP)	25	4	86.2	87.1	86.2	87.1	6.6 (2.6–16.8)
II (PP, NA, AP)	4	16	51.6	86.2	80.0	62.5	3.7 (1.4–9.8)
III (AA)	0	11	35.5	100	100	59.2	9.9 (1.3–73.3)

PPV, positive predictive value; NPV, negative predictive value; LR+, likelihood ratio; and CI, confidence interval at 95% (in bold type cases correctly classified).

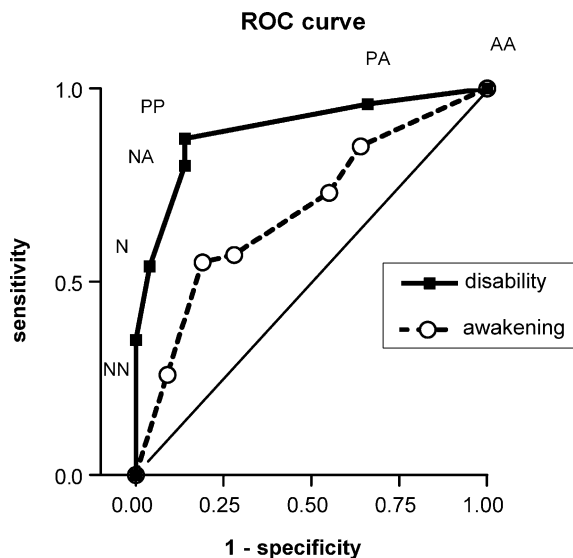


Fig. 2. Receiver operator characteristics (ROC) curves showing accuracy for prediction of both functional outcome (disability) and 'awakening' in post-traumatic coma according to SEP findings. The ordinate axis shows the sensitivity of the tests, ranging from 0 to 1.0 (0–100%), while the abscissa shows the percentage of false positive results (1-specificity). Tests with good discriminatory power produce a ROC curve, which closely follows the left hand axis and the top margin of the graph, passing close to the upper left corner. Such tests have both high sensitivity and high specificity.

predicting awakening and good outcome. But, because of a different sensitivity, the positive likelihood ratio was greater ($LR+ = 6.6$) in predicting the level of disability, than 'awakening' ($LR+ = 4.5$). SEP grade III is a powerful predictor for both 'awakening' ($LR+ = 8.0$) and, above all, bad outcome ($LR+ = 9.9$). SEP grade II is associated with the wider range of outcome and its predictive value depends on the outcome categorization ($LR+$ was, respectively, one for 'awakening' and 3.7 for disability).

Compared with SEPs, GCS and EEG reactivity showed a lower sensitivity, specificity and predicting value (Tables 2 and 3). Indeed, the univariate analysis reveals that SEPs have been able to correctly classify a higher percentage of patients. 'Awakening' or death/PVS were correctly predicted in 83.3, 65.0 and 71.6%, respectively, for SEPs, EEG reactivity and GCS; disability in 86.6, 68.3 and 68.3%. A multivariate analysis including all 3 predictors did not increase the percentage of cases correctly predicted with SEPs alone. Nevertheless, EEG was useful in the SEP grade II: all the 10 patients with unreactive EEGs had a bad outcome.

4. Discussion

Systematic reviews of the literature have provided evidence of the usefulness of SEPs in predicting the outcome of post-traumatic coma (Attia and Cook, 1998; Carter and Butt, 2001; Robinson et al., 2003; Rothstein, 2000). However, it is difficult to compare the individual

studies because of the different patient selection, SEP recording time and grading criteria (Anderson et al., 1984; Brown et al., 1998; Cant et al., 1986; Claassen and Hansen, 2001; Facco et al., 1990; Firsching et al., 1998; Greenberg et al., 1977; Guérit et al., 1993; Gütlings et al., 1993, 1994; Hutchinson et al., 1991; Logi et al., 2003; Rumpl et al., 1983; Sleight et al., 1999).

In our study, the ROC curves indicated that the most powerful aggregation of SEP patterns able to reliably predict the outcome for as many patients as possible, resulted to be a 3-graded scale: grade I (NN, NP) as a favourable pattern and grade III (AA) as an unfavourable pattern. The predictive value of grade II (AN, AP, PP) was lower and it was dependent on the outcome categorization.

Patients with SEP grade I have a high chance of regaining consciousness (PPV 93.1%) and of surviving with an acceptable quality of life (PPV 86.2%). Exceptions to this clear relationship deserve a brief discussion. An initial preserved bilateral cortical response does not imply in all cases a good prognosis, as shown in those 2 patients who died without regaining consciousness (one from non-neurological causes). Some studies report an increased prognostic reliability of SEPs with serial recordings for at least 2 weeks (Pohlmann-Eden et al., 1997), thus we can assume that our timing of SEP recording greatly reduced, but not totally avoided the chance of false optimistic prediction. On the other hand, hypoxic damage can occur in addition to the traumatic insult thus reducing the favourable predictive value of a preserved evoked response on primary parietal areas. All patients with SEP grade III had a bad outcome (GOS 1–3, PPV = 100%). This agrees with the literature, according to which there are only exceptional cases of patients with a bilaterally absent SEPs who recover with a good outcome (GOS 4–5) (Rumpl et al., 1983; Tozzi et al., 1990). The percentage of our patients with SEP grade III who regained consciousness, but were left with severe disability, was significantly greater (27%) than reported in the literature (10% in Rothstein (2000) and 5% in Robinson et al. (2003)). This discrepancy could be due to different reasons. The reappearance of a bilaterally absent SEP is possible, even if extremely rare (22 patients over 2632 cases, Carter and Butt, 2001) and it is usually observed within 10 days from the injury (Claassen and Hansen, 2001), thus with our testing time (7–10 days) we should have avoided this possibility. Moreover, none of the 3 patients showed the conditions (recent craniotomy, subdural or extradural collections and focal lesions preventing impulses to reach cortex) considered a possible source of unfavourable prognostic error (Carter and Butt, 2001). It is more likely, instead, that such a discrepancy could be due to our inclusion criteria and/or follow-up duration. Indeed, having selected patients with a long duration of acute coma, a prolonged state of unconsciousness was expected. Most of the studies in the meta-analysis have a period of follow-up of 3–6 and, more rarely 12 months, whereas we evaluated the outcome after at least 12 months. This longer timing of

follow-up may explain our higher percentage of patients SEP grade III who ‘awoke’ considering that they recovered a clinical responsiveness, respectively, at 3, 6 and 8 months.

SEP grade II has a lower predictive value compared to SEP grades I and III and it depends on the outcome categorization, being usually favourable for ‘awakening’ and unfavourable for disability.

On the whole, with our SEP grading, we can correctly classify up to 83.3 and 86.6% of cases, respectively, in the category of ‘awakening’ and disability.

An extensive review of the literature reveals that neither GCS nor EEG are as helpful in traumatic as in hypoxic–ischemic coma (Attia and Cook, 1998; Facco and Munari, 2000). Concerning EEG, in post traumatic coma a predictive value is usually ascribed to the presence of reactivity (Fischer and Mutschler, 2002), and, according to some authors, its predictive value may be greater than that of SEPs (Gütling et al., 1995). In our study, GCS and EEG reactivity after 1 week, separately considered, have shown a lower predictive index (PPV, LR+) compared to SEP grading. Including in a multivariate analysis SEP grading, EEG reactivity and GCS we have not increased the number of patients correctly predicted using SEPs alone. Furthermore, the analysis of neurophysiological data in individual patients with SEP grades I and III has revealed that, in case of discordance between SEPs and EEG, the prognosis was always in line with the SEP pattern. More useful, instead, is the role of EEG in SEP grade II: an unreactive EEG in these patients always predicts a bad outcome.

In our analysis, we have not carried out a prognostic comparison with neuroradiological data, but recent studies assert that conventional MRI does not improve the prognostic reliability of SEPs in head injury, but rather offers possibilities for clarifying neurophysiological and clinical pathologies (Soldner et al., 2001).

To date, most analysis of post-traumatic coma have focused on using SEPs to predict ‘non-awakening’ rather than good outcome. This might be due to the fact that meta-analysis obliges an excessive simplification of both prognostic categories and SEP findings. Probably, the inclusion of normal and pathological responses in the same category of ‘preserved’ SEPs has led, in many studies, to a reduction of its favourable predictive value (Rothstein, 2000). Few authors (Fischer and Mutschler, 2002; Guérit, 1999) have indeed underlined the meaning of normal cortical SEPs as an index of preserved cortical function and so its useful prognostic value for predicting good outcome. Only the most recent meta-analysis (Robinson et al., 2003) reports a high favourable PPV of normal SEPs both for ‘awakening’ (89%) and recovery according to GOS (79%).

In our study, the presence of normal SEPs (grade I) has a favourable predicting value of more than 90% for ‘awakening’ and 80% for functional recovery confirming that SEPs maintain a high prognostic value even in the category of TBI that we have selected according to the severity and long duration of coma (>7 days). This is

a consistent subcategory of patients, ~40% according to Facco and Giron (2002), whose prognosis is particularly problematic. In post-traumatic coma, the absence of a selective intracortical vulnerability, typical of anoxic–hypoxic damage, and the importance of DAI (diffuse axonal injury) on the brainstem and thalamo-cortical projections make SEPs a good index of global cerebral function, even if we keep in mind that we are testing a very limited sample of the nervous system.

The functional state of a widespread cortical associative area is explored by event-related potentials (ERPs), therefore there is a growing interest for their applicability in the prediction of ‘awakening’ in comatose patients. There is preliminary evidence (Fischer et al., 1999; Guérit et al., 1999; Kane et al., 1996) that the presence of any recognizable cognitive potential (MMN, P300) has a very high PPV for the recovery of a conscious state. There is only one recent report that has compared (Lew et al., 2003) the predictive value of SEPs and speech-evoked ERPs showing for the latter a superior PPV for a good outcome. We agree with Facco and Giron (2000) that short latency SEPs maintain their value in the acute phase of coma, while ERPs add information in the subacute phase in those patients still unresponsive and with uncertain possibility of recovery.

Prognostication of severe TBI is based on the integration of clinical, neuroradiological and electrophysiological features by an expert intensivist and neurophysiologist. Lacking a SEP pattern with absolute predictive value, the prediction for individual patients is still not reliable enough to modify the clinical management, but SEPs provide early information on the expectation of recovery. SEP testing is a simple technique, non-expensive, reliably recorded at the bedside and its ability to predict both good and bad outcome in severely head injured patients is superior to GCS and EEG reactivity. So, this technique should be used more widely in the assessment of coma.

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